

Q: find all divisors of 24 (num).

$$24 \rightarrow 1, 2, 3, 4, 6, 8, 12, 24.$$

$$\frac{24}{1} \% 1 == 0$$

$$24 \% 2 == 0$$

$$\rightarrow 2$$

$$\frac{24}{3} \rightarrow 3$$

$$\frac{24}{4} \rightarrow 4$$

$$\frac{24}{1} \checkmark \quad \frac{24}{2} \checkmark$$

$$1, 2, 3, \dots, 24 \quad \frac{24}{3} \checkmark$$

$$\text{num} = 24$$

$$\text{num}$$

for (int i = 1; i <= 24; i++) {

if (num % i == 0) cout << i; ✓

}

$$1 \rightarrow n$$

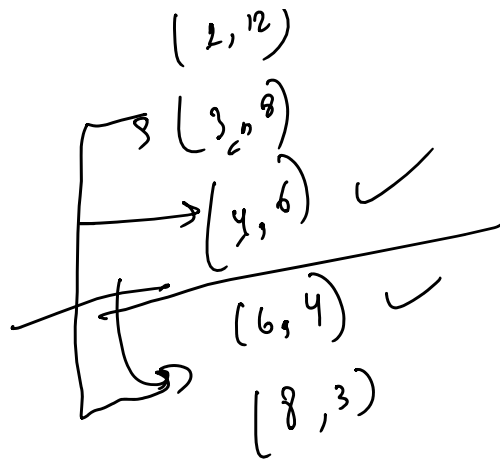
$$\frac{24}{1}, \quad \textcircled{24}$$

$$\begin{pmatrix} 1, 24 \\ 2, 12 \\ 3, 8 \\ 4, 6 \end{pmatrix}$$

$$24 = 12$$

$$\frac{24}{2} = 12$$

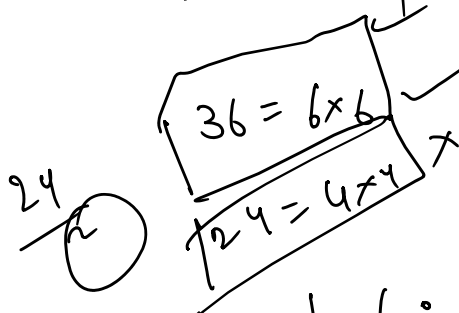
$$\frac{24}{6} = 4$$



$$4 \times 4 = 16$$

$$5 \times 5 = 25$$

$$\sqrt{24} = 4$$



int 4

$n = 24$, 28

for (int i = 1; i <= $\sqrt{n}$; i++) {

if (n % i == 0) {

→ cout << i;

if (n/i != i) cout << (n/i);

i = 6
n = 36

$$\frac{36}{6} = 6$$

2, 12
4, 6

$n = 36 \rightarrow 1, 36$

2, 18

3, 12

4, 9

6, 6

9, 4

12, 3

$$\sqrt{36} = 6$$

i = 1 n = 36

$$l = 1, n = 36$$

$$\frac{36}{6} = 6$$

$$\frac{36}{6} = 6$$

How find out the number of divisions;

Q. Find the HCF of two numbers.

$$\hookrightarrow \text{HCF} (9, 34)$$

$\hookrightarrow$ Highest common factor

$$\text{HCF} = 1$$

$$9 \rightarrow 1, 3, 9$$

$$34 \rightarrow 1, 2, 17, 34$$

$$(25, 35)$$

$$\hookrightarrow 25 - 1, 5, 25$$

$$35 - 1, 5, 7, 35$$

$$\text{HCF} = 5$$

$$(25, 35)$$

$$\text{HCF} = 5$$

$$1, 2, 3, \dots, 5, \dots, 25$$

$$\frac{35}{7} \checkmark$$

$$\frac{25}{5} \checkmark$$

$$\frac{35}{5}, \frac{25}{5}$$

$$\frac{35}{5}, \frac{25}{5}$$

(35) (25)

35/1, 25/1

100, 100 0000

25, 35
→

$\frac{25}{26}$ 26

int a = 25, b = 35;

int Hcf = 1;

for (int i = 2; ^{Math.} min(a, b); i++) {
if (a % i == 0 && b % i == 0) Hcf = i;
}

}